

API Documentation

Eila Eco Pencils E-Commerce Platform

1. Introduction

This document defines the backend APIs for the Eila Eco Pencils platform. The frontend communicates with Supabase for CRUD operations while the Node.js backend exposes secure endpoints for payment verification, analytics, and server-side logic.

Architecture

REST APIs over HTTPS using JSON. JWT-based admin authentication through Supabase Auth. Razorpay verification handled only on the backend.

Authentication

Admins authenticate with Supabase Auth. Public endpoints expose only active catalog data. Protected endpoints require a valid bearer token.

API Standards

Use HTTPS, JSON payloads, camelCase field names, semantic HTTP methods, versioning (/api/v1), and consistent error responses.

Endpoint Catalog

Method	Endpoint	Purpose
GET	/api/v1/products	List active products
GET	/api/v1/products/:slug	Get product details
GET	/api/v1/categories	List categories
POST	/api/v1/payment/create-order	Create Razorpay order
POST	/api/v1/payment/verify	Verify payment signature
GET	/api/v1/admin/orders	List orders
PATCH	/api/v1/admin/orders/:id	Update order status
POST	/api/v1/admin/products	Create product
PUT	/api/v1/admin/products/:id	Update product
DELETE	/api/v1/admin/products/:id	Delete product
GET	/api/v1/admin/analytics	Dashboard metrics

Request & Response Format

All requests use application/json. Successful responses include success=true and data. Errors include success=false, message, and optional validation details.

Implementation Notes

Document request parameters, response schemas, edge cases, authorization requirements, performance considerations, and example payloads for each endpoint. Document request parameters, response schemas, edge cases, authorization requirements, performance considerations, and example payloads for each endpoint. Document request parameters, response schemas, edge cases, authorization requirements, performance considerations, and example payloads for each endpoint.

HTTP Status Codes

200 OK, 201 Created, 204 No Content, 400 Bad Request, 401 Unauthorized, 403 Forbidden, 404 Not Found, 409 Conflict, 422 Validation Error, 500 Internal Server Error.

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Validation Rules

Validate product IDs, stock values, email addresses, phone numbers, payment signatures, and required checkout fields before processing.

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Security

Rate limiting, CORS, input sanitization, environment variables, HTTPS, JWT validation, and backend-only Razorpay secret usage.

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Error Handling

Provide descriptive messages, log server failures, avoid exposing internal details, and return consistent JSON structures.

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Versioning

All endpoints are prefixed with /api/v1. Breaking changes should be introduced through new API versions.

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Testing

Validate each endpoint using Postman, automated integration tests, payment sandbox testing, and negative test scenarios.

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Future APIs

Wishlist, reviews, coupons, invoices, AI recommendations, mobile APIs, and order tracking notifications.

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